Calculus and Linear Algebra

# Fermat, Rolle and Mean Value Theorems

Lecture 20

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# Lecture outline

- Local maxima and minima.
- Fermat's theorem: necessary condition at an interior extremum.
- Rolle's theorem and proof.
- Lagrange's Mean Value Theorem and proof from Rolle.
- Consequences for monotonicity and constant functions.
- Cauchy's Mean Value Theorem.
- Examples, counterexamples and exercises.

**Main theme**

Local information about derivatives can force global conclusions about a function on an interval.

# Local extrema

**Definition** A function $f$ has a **local maximum** at $x_0$ if there exists a neighbourhood of $x_0$ such that

$$f(x) \leq f(x_0)$$

for all $x$ in that neighbourhood. Similarly, $f$ has a **local minimum** at $x_0$ if

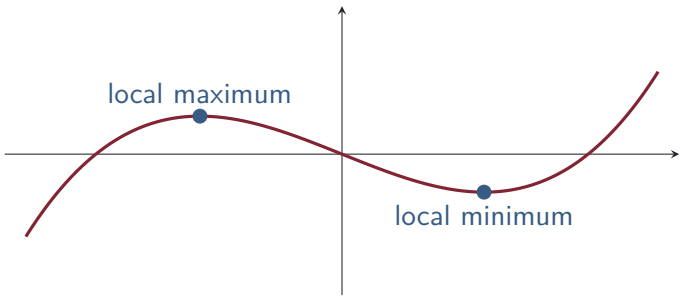
$$f(x) \geq f(x_0)$$

near $x_0$.

**Local means nearby**

A local extremum need not be the largest or smallest value on the whole domain.

# Local versus absolute extrema




The derivative will help us detect such points, but first we need a theorem.

# Fermat's theorem

**Theorem** Suppose $f$ has a local maximum or minimum at an **interior point** $x_0$ and is differentiable at $x_0$. Then

$$\boxed{f'(x_0) = 0.}$$

## Geometric meaning

At a smooth interior extremum, the tangent line is horizontal.

This is a **necessary condition**, not a sufficient one.

# Why Fermat is plausible

At a local maximum $x_0$, for small $h > 0$,

$$f(x_0 + h) - f(x_0) \leq 0,$$

so

$$\frac{f(x_0 + h) - f(x_0)}{h} \leq 0.$$

For small $h < 0$, the denominator changes sign, hence

$$\frac{f(x_0 + h) - f(x_0)}{h} \geq 0.$$

If the two one-sided derivative limits agree, their common value must be

$$\boxed{0.}$$

The minimum case is analogous.

# Critical points

A point $x_0$ in the domain is called a **critical point** if

$$f'(x_0) = 0$$

or $f'(x_0)$ does not exist. Fermat says that an interior differentiable local extremum must occur among critical points.

**But be careful**

Critical points are only **candidates** for extrema.

# Stationary point does not imply extremum

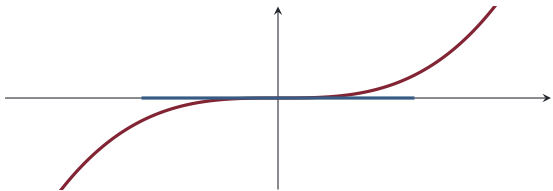
Consider

$$f(x) = x^3.$$

Then

$$f'(x) = 3x^2, \qquad f'(0) = 0.$$

But $x = 0$ is neither a local maximum nor a local minimum.



# Rolle's theorem

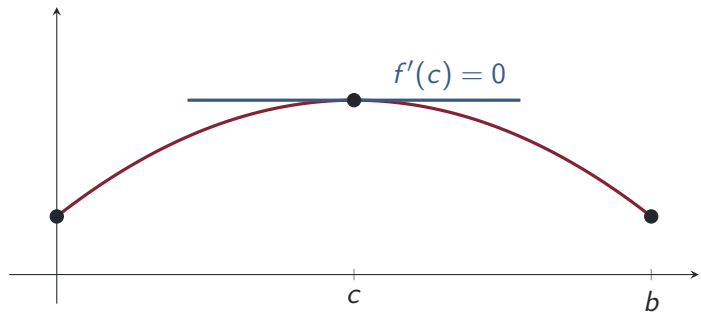
**Theorem** Let $f$ satisfy:

1. $f$ is continuous on $[a, b]$;
2. $f$ is differentiable on $(a, b)$;
3. $f(a) = f(b)$.

Then there exists at least one $c \in (a, b)$ such that

$$\boxed{f'(c) = 0.}$$

# Rolle's theorem: geometric picture




Equal endpoint values force at least one horizontal tangent, provided continuity and differentiability hold.

# Proof of Rolle's theorem

Because $f$ is continuous on $[a, b]$, the Extreme Value Theorem guarantees an absolute maximum and minimum. If both are equal, $f$ is constant and every interior point satisfies $f'(c) = 0$. Otherwise at least one absolute extremum occurs at an interior point because

$$f(a) = f(b).$$

At that interior extremum, Fermat's theorem gives

$$\boxed{f'(c) = 0.}$$

# Rolle: a concrete example

Let

$$f(x) = x^2 - 4x + 3$$

on $[1, 3]$.

$$f(1) = 0, \qquad f(3) = 0.$$

The hypotheses of Rolle are satisfied. Since

$$f'(x) = 2x - 4,$$

we solve

$$2c - 4 = 0$$

and obtain

$$\boxed{c = 2.}$$

# Why the hypotheses matter

Each assumption in Rolle's theorem has a role.

- Without continuity on $[a, b]$, the graph may jump.
- Without differentiability on $(a, b)$, a corner can replace the horizontal tangent.
- Without $f(a) = f(b)$, there is no reason for a horizontal tangent to occur.

**Theorem use**

Never quote a theorem before checking its hypotheses.

# Counterexample: differentiability is necessary

Consider

$$f(x) = |x|$$

on $[-1, 1]$.

$$f(-1) = f(1) = 1.$$

The function is continuous, but it is not differentiable at 0. There is no point in $(-1, 1)$ where

$$f'(x) = 0.$$

So Rolle's conclusion fails when differentiability fails.

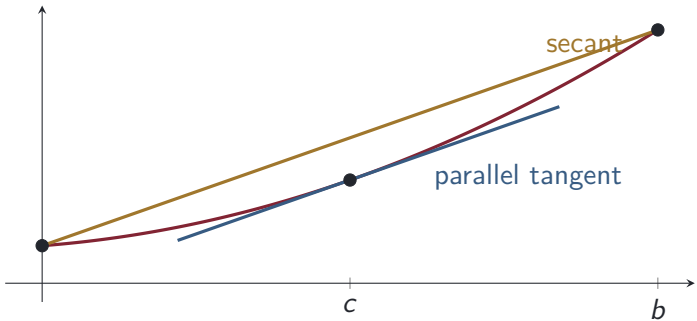
# Lagrange's Mean Value Theorem

**Theorem** If $f$ is continuous on $[a, b]$ and differentiable on $(a, b)$, then there exists at least one $c \in (a, b)$ such that

$$\boxed{f'(c) = \frac{f(b) - f(a)}{b - a}.}$$

The derivative at some interior point equals the average rate of change over the whole interval.

# MVT: geometric interpretation




At some point, the tangent is parallel to the secant joining the endpoints.

# MVT example: $f(x) = x^2$ on $[1, 3]$

The average rate of change is

$$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4.$$

Since

$$f'(x) = 2x,$$

we seek $c$ such that

$$2c = 4.$$

Thus

$$\boxed{c = 2.}$$

# Proof of the Mean Value Theorem

Let $\ell(x)$ be the secant line through $(a, f(a))$ and $(b, f(b))$:

$$\ell(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

Define

$$g(x) = f(x) - \ell(x).$$

Then

$$g(a) = g(b) = 0.$$

By Rolle's theorem, there exists $c \in (a, b)$ such that

$$g'(c) = 0.$$

# Completing the proof

Since

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

Rolle gives

$$0 = g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

Therefore

$$\boxed{f'(c) = \frac{f(b) - f(a)}{b - a}.}$$

The MVT is thus a direct consequence of Rolle applied after subtracting the secant line.

# First consequence: derivative zero everywhere

Suppose

$$f'(x) = 0$$

for every $x$ in an interval $I$. Take any $a < b$ in $I$. By the MVT, for some $c \in (a, b)$,

$$\frac{f(b) - f(a)}{b - a} = f'(c) = 0.$$

Hence

$$f(b) = f(a).$$

Therefore

$$\boxed{f \text{ is constant on } I.}$$

# Second consequence: positive derivative

Suppose

$$f'(x) > 0$$

for every $x$ in an interval $I$. For $a < b$, the MVT gives

$$\frac{f(b) - f(a)}{b - a} = f'(c) > 0.$$

Since $b - a > 0$,

$$f(b) - f(a) > 0.$$

Thus

$$\boxed{f \text{ is strictly increasing on } I.}$$

# Third consequence: negative derivative

Analogously, if

$$f'(x) < 0$$

throughout an interval $I$, then for $a < b$,

$$\frac{f(b) - f(a)}{b - a} = f'(c) < 0.$$

Hence

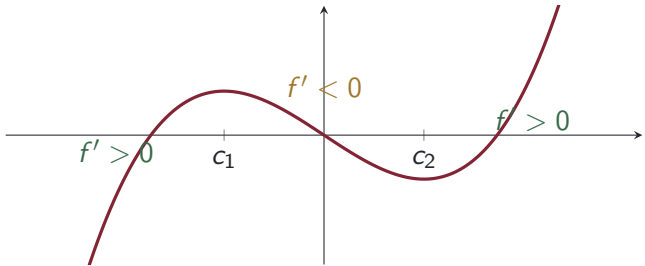
$$f(b) < f(a).$$

Therefore

$$\boxed{f \text{ is strictly decreasing on } I.}$$

These results justify derivative sign charts.

# Derivative sign and graph behaviour




The MVT turns the sign of the derivative into a rigorous statement about monotonicity.

# Exercise: verify the MVT hypotheses

**Exercise** Consider

$$f(x) = \sqrt{x}$$

on $[1, 4]$.

1. Is $f$ continuous on $[1, 4]$?
2. Is it differentiable on $(1, 4)$?
3. Find the value $c$ guaranteed by the MVT.

# Exercise: solution

**Solution** The hypotheses are satisfied.

$$\frac{f(4)-f(1)}{4-1}=\frac{2-1}{3}=\frac{1}{3}.$$

Since

$$f'(x)=\frac{1}{2\sqrt{x}},$$

we solve

$$\frac{1}{2\sqrt{c}}=\frac{1}{3}.$$

Hence

$$2\sqrt{c}=3, \qquad \boxed{c=\frac{9}{4}.}$$

# Cauchy's Mean Value Theorem

**Theorem** Let $f$ and $g$ be continuous on $[a, b]$ and differentiable on $(a, b)$. Then there exists $c \in (a, b)$ such that

$$\boxed{[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).}$$

If additionally $g(b) \neq g(a)$ and $g'(c) \neq 0$, this may be written as

$$\boxed{\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.}$$

# Lagrange as a special case of Cauchy

Choose

$$g(x) = x.$$

Then

$$g'(x) = 1, \qquad g(b) - g(a) = b - a.$$

Cauchy's theorem becomes

$$\boxed{f'(c) = \frac{f(b) - f(a)}{b - a},}$$

which is exactly Lagrange's Mean Value Theorem.

# Proof idea for Cauchy's theorem

Define

$$H(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x).$$

Then

$$H(b) - H(a) = 0,$$

so

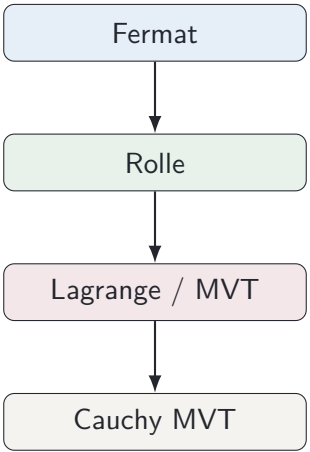
$$H(a) = H(b).$$

Rolle's theorem gives some $c \in (a, b)$ with

$$H'(c) = 0.$$

Expanding $H'(c) = 0$ gives Cauchy's relation.

# The hierarchy of theorems




Each theorem generalises or builds on the previous ideas.

# Existence is not uniqueness

Rolle and MVT guarantee **at least one** point $c$. They do not say that $c$ is unique. For example, a function may have several horizontal tangents while satisfying

$$f(a) = f(b).$$

**Important logical distinction**

$$\exists c$$

does not mean

$$\exists! c.$$

# When a theorem cannot be applied

**Exercise** Can the MVT be applied to

$$f(x) = |x|$$

on $[-1, 1]$? The function is continuous on $[-1, 1]$, but it is not differentiable at $0 \in (-1, 1)$. Therefore the hypotheses fail.

The MVT cannot be applied.

This does not mean every possible conclusion is false; it means the theorem does not guarantee it.

# Exercise: Fermat or not?

**Exercise** For each statement, decide whether Fermat's theorem applies.

1. $f(x) = x^2$ at $x = 0$.
2. $f(x) = |x|$ at $x = 0$.
3. $f(x) = x^3$ at $x = 0$.
4. $f(x) = x$ at the left endpoint $x = 0$ of $[0, 1]$.

Focus on the hypotheses, not only on the graph.

# Exercise: answers

**Solution**

1. Yes: differentiable interior minimum, and $f'(0) = 0$.
2. No: local minimum but not differentiable at 0.
3. Fermat's conclusion $f'(0) = 0$ holds, but there is no extremum; the converse is false.
4. No: 0 is not an interior point of the interval.

# Exercise: monotonicity from a derivative

Suppose

$$f'(x) = 3x^2 + 1.$$

Since

$$3x^2 + 1 > 0$$

for every $x \in \mathbb{R}$, the MVT consequence gives

$$\boxed{f \text{ is strictly increasing on } \mathbb{R}.}$$

No explicit formula for $f$ is needed to deduce monotonicity.

# Exercise: another derivative sign

Suppose

$$f'(x) = x^2 - 4 = (x-2)(x+2).$$

Then

$$f'(x) > 0 \quad \text{for } x < -2 \text{ or } x > 2,$$

$$f'(x) < 0 \quad \text{for } -2 < x < 2.$$

Therefore $f$ is increasing, then decreasing, then increasing. This is exactly the type of analysis we will systematise next.

# The theorem checklist

Before applying Rolle, Lagrange or Cauchy, ask:

1. What is the interval?
2. Is the function continuous on the **closed** interval?
3. Is it differentiable on the **open** interval?
4. Are there extra hypotheses, such as $f(a) = f(b)$ for Rolle?
5. What equation must the point $c$ satisfy?

**Method matters**

The hypotheses are part of the solution, not a formality.

# Independent practice

**Exercise**

1. Apply Rolle to $f(x) = x^2 - 5x + 6$ on $[2, 3]$.
2. Apply the MVT to $f(x) = x^3$ on $[0, 2]$ and find all possible $c$.
3. Show that if $f'(x) = 2$ on an interval, then $f(x) - 2x$ is constant.
4. Decide whether the MVT applies to $f(x) = 1/x$ on $[-1, 1]$.
5. State Cauchy's theorem and explain how Lagrange follows from it.

# Independent practice: short answers

**Solution**

1. $f(2) = f(3) = 0$ and $f'(x) = 2x - 5$, so $c = 5/2$.
2. Average slope is 4; solve $3c^2 = 4$, giving $c = 2/\sqrt{3}$ in $(0, 2)$.
3. $(f(x) - 2x)' = 0$, hence it is constant by the MVT consequence.
4. No: $1/x$ is not continuous on $[-1, 1]$.
5. Take $g(x) = x$ in Cauchy's theorem.

# What we know now

- Fermat identifies a necessary condition for smooth interior extrema.
- Rolle guarantees a horizontal tangent when endpoint values coincide.
- Lagrange connects instantaneous and average rates of change.
- The MVT justifies monotonicity tests based on the sign of $f'$.
- Cauchy generalises the MVT to two functions.

# Looking ahead

Next we use these theorems operationally. Given $f$, we will:

$$\boxed{\text{compute } f' \longrightarrow \text{ study its sign } \longrightarrow \text{ deduce monotonicity.}}$$

Then we will identify

local maxima, local minima and stationary points,

and begin assembling the full procedure for a complete function study.